

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 4: Distance Measuring Equipment & Slant Range Geometry

Syllabus Source: CAE Oxford (pp. 131-168) | Pass Target: $\geq 90\%$ AviationExam

1. DME OPERATING PRINCIPLES & UHF FREQUENCIES

DME operates in the UHF band from 960 to 1215 MHz (1 MHz spacing). It is a secondary radar system measuring line-of-sight distance:

- **Pulse Pair Interrogation:** Airborne interrogator transmits pulse pairs (12 microsecond spacing). Ground transponder receives them, delays reply by exactly 50 microseconds, and re-transmits on a frequency offset by +63 MHz.
- **Slant Range Formula:** Total Elapsed Time = Round-trip propagation time + 50 microsecond transponder delay. Distance (NM) = (Elapsed Time - 50 microsec) / 12.36 microseconds (since radio waves travel 1 NM round-trip in 12.36 microseconds).
- **Capacity & Saturation:** A DME ground beacon can handle up to 100 AIRCRAFT SIMULTANEOUSLY. If overloaded, it reduces receiver sensitivity, cutting off weaker (distant) interrogators.

2. SLANT RANGE ERROR (OVERHEAD GEOMETRY)

DME measures direct line-of-sight hypotenuse distance (Slant Range), NOT horizontal ground distance:

- **The Pythagoras Geometry:** Ground Distance = $\sqrt{\text{Slant Range}^2 - \text{Height}^2}$ [Height in NM: Height (NM) = Altitude in feet / 6,076 ft].
- **Overhead Station Indication:** When passing DIRECTLY OVERHEAD the DME station, Ground Distance is ZERO, but the DME indicates the aircraft's ALTITUDE ABOVE THE STATION IN NAUTICAL MILES (e.g. at FL 360, DME reads 6.0 NM overhead!)

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Evaluating DME Slant Range Error

Problem: Cruising at FL 360 (36,000 ft) toward a VOR/DME. DME indicator reads 10.0 NM. What is the actual horizontal ground distance to the station? Step 1: Convert aircraft altitude to nautical miles: $36,000 \text{ ft} / 6,076 \text{ ft/NM} = 5.925 \text{ NM}$ (approx 6.0 NM). Step 2: Apply Pythagorean theorem: $\text{Ground Dist}^2 = \text{Slant Range}^2 - \text{Height}^2$. Step 3: $\text{Ground Dist}^2 = 10.0^2 - 6.0^2 = 100 - 36 = 64 \text{ NM}^2$. Step 4: Ground Distance = $\sqrt{64} = 8.0 \text{ NM}$! Conclusion: DME overreads by 2.0 NM due to slant range. Error becomes negligible when distance $> 3 \times$ altitude in thousands.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating True Ground Distance from DME Slant Range

SCENARIO (Standard EASA DME Geometry Problem): An airliner is descending at FL 240 (24,000 ft above a sea-level DME station): - The cockpit DME indicator reads 5.0

NM QUESTION: What is the aircraft's actual HORIZONTAL GROUND DISTANCE from the station? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Convert Aircraft Height from

feet into Nautical Miles: - Rule: 1 Nautical Mile = approx 6,000 feet (exact 6,076 ft). - Height (NM) = 24,000 ft / 6,000 ft = 4.0 NM! Step 2: Draw the Right-Angled

Triangle: - Hypotenuse (DME Slant Range) = 5.0 NM - Vertical leg (Height) = 4.0

NM - Horizontal leg (Ground Distance) = ? Step 3: Apply Pythagoras' Theorem ($A^2 + B^2 = C^2$): - Ground Distance = $\sqrt{\text{Slant Range}^2 - \text{Height}^2}$ - Ground Distance

= $\sqrt{5.0^2 - 4.0^2} = \sqrt{25.0 - 16.0} = \sqrt{9.0} = 3.0$ NM! FINAL ANSWER:

The actual horizontal ground distance to the station is ONLY 3.0 Nautical Miles (even though the DME reads 5.0 NM!).

[!] EXAM TRAP (AVIATIONEXAM) - The Slant Range Error Vanishes at Distance Trap

- When Slant Range is greater than 10 times the height (e.g. 40 NM range at 4 NM height): - Slant range error is less than 0.5% (negligible)! - Slant range error is ONLY dangerous close to or directly overhead the station.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What will a DME indicator read when an aircraft is flying directly overhead a sea-level DME beacon at FL 360?

- [A]: Exactly 0.0 NM
- [B]: APPROXIMATELY 6.0 NM (36,000 ft / 6,000 ft/NM = 6.0 NM slant range)
- [C]: 36 NM
- [D]: The indicator displays dashes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Because DME measures line-of-sight distance, passing overhead produces a reading equal to the aircraft's vertical altitude in Nautical Miles: 36,000 / 6,000 = 6.0 NM.

Q2: In which frequency band does Distance Measuring Equipment (DME) operate?

- [A]: VHF
- [B]: UHF (960 to 1215 MHz)
- [C]: SHF
- [D]: HF

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. DME operates in the UHF band, paired with VOR channels so tuning the VOR frequency automatically tunes the DME interrogator.

Q3: What is the maximum number of aircraft that a standard DME ground transponder can track simultaneously?

- [A]: Unlimited
- [B]: APPROXIMATELY 100 AIRCRAFT
- [C]: 25 aircraft
- [D]: 500 aircraft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. DME ground transponders are designed to handle up to 100 aircraft interrogations simultaneously by limiting pulse reply rates to ~2,700 pulse pairs per second.